

Online Ticket Reservation and Seat Locking System

Database Tables:

1) USERS Table

Stores Customer, Admin, and Service_provider details.

Column	Type	Constraints	Purpose
user_id	INT	PK, AUTO_INCREMENT, NOT NULL	Unique ID for each user
name	VARCHAR (100)	NOT NULL	User's name
email	VARCHAR (150)	NOT NULL, UNIQUE	Login and communication
phone	VARCHAR (15)	UNIQUE	Contact number
password	VARCHAR (255)	NOT NULL	Stores password hash
role	VARCHAR (30)	NOT NULL	CUSTOMER/ ADMIN/ SERVICE_PROVIDER
provider_name	VARCHAR (150)	NULL	Company/theatre business name
provider_type	VARCHAR (50)	NULL	Theatre Owner/ Movie Distributor
business_address	VARCHAR (255)	NULL	Provider's business address
business_license	VARCHAR (100)	NULL	Business registration/ License number
status	VARCHAR (20)	DEFAULT 'ACTIVE'	Account status
verification_status	VARCHAR (20)	DEFAULT 'NOT_REQUIRED'	Provider verification status
verified_at	DATETIME	NULL	Date/time provider was verified
created_at	DATETIME	DEFAULT CURRENT_TIMESTAMP	Account creation time

2) MOVIES Table

Stores movie and show-related information.

Column	Type	Constraints	Purpose
movie_id	INT	PK, AUTO_INCREMENT, NOT NULL	Unique movie ID
title	VARCHAR (150)	NOT NULL	Movie name
language	VARCHAR (50)	NULL	Movie description
genre	VARCHAR (50)	NOT NULL	Action, Comedy, Drama etc.
duration_minutes	INT	NOT NULL	Movie duration time
release_date	DATE	NULL	Movie release date
theatre_name	VARCHAR (150)	NOT NULL	Theatre where movie is shown
location	VARCHAR (225)	NOT NULL	Theatre location
show_date	DATE	NOT NULL	Date of show
start_time	TIME	NOT NULL	Show starting time
end_time	TIME	NOT NULL	Show ending time
ticket_price	DECIMAL (10,2)	NOT NULL	Price per ticket
status	VARCHAR (20)	DEFAULT 'ACTIVE'	Movie/show status
created_by	INT	FK → USERS	Admin/provider who added it
Created_at	DATETIME	DEFAULT CURRENT_TIMESTAMP	Creation date

3) SEATS Table

This table manages seat details + seat locking

Column	Type	Constraints	Purpose
seat_id	INT	PK, AUTO_INCREMENT, NOT NULL	Unique seat ID
Movie_id	INT	FK → MOVIES, NOT NULL	Identifies movie/show
seat_number	VARCHAR (20)	NOT NULL	A1, A2, B1 ECT.
seat_type	VARCHAR (30)	NOT NULL	REGULAR/ PREMIUM/ VIP
status	VARCHAR (20)	DEFAULT 'AVAILABLE'	Current seat state
locked_by	INT	FK → USERS, NULL	User currently locking seat
locked_at	DATETIME	NULL	Lock start time
expires_at	DATETIME	NULL	Lock expires time

4) BOOKINGS Table

Stores the customer's ticket reservation.

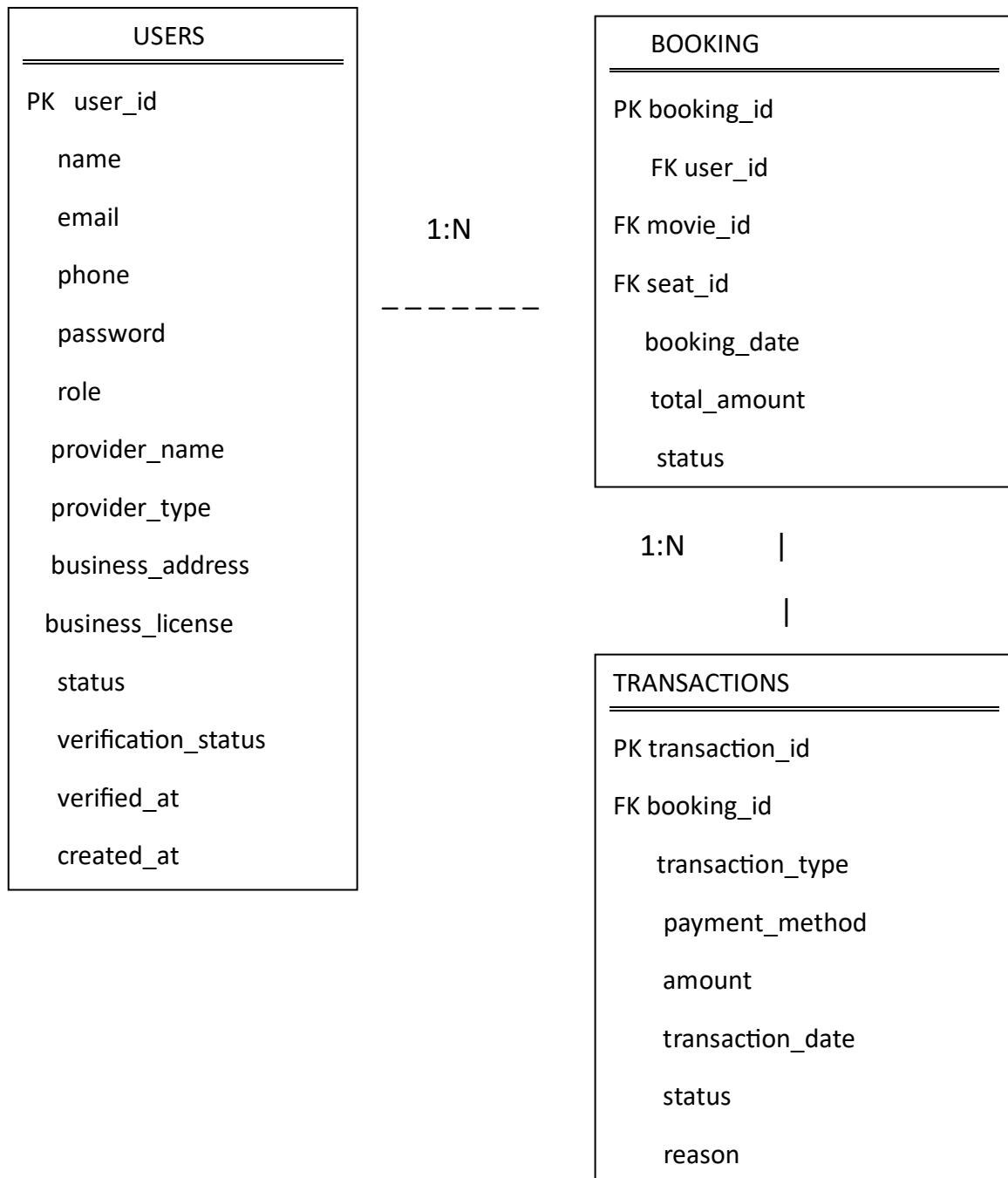
Column	Type	Constraints	purpose
booking_id	INT	PK, AUTO_INCREMENT, NOT NULL	Unique booking ID
user_id	INT	FK → USERS, NOT NULL	Customer who booked
movie_id	INT	FK → MOVIES, NOT NULL	Movie being booked
seat_id	INT	FK → SEATS, NOT NULL	Seat being booked
booking_date	DATETIME	DEFAULT CURRENT_TIMESTAMP	Booking date/time
total_amount	DECIMAL (10,2)	NOT NULL	Total booking amount
status	VARCHAR (20)	DEFAULT 'PENDING'	Booking status

5) TRANSACTIONS Table

Stores payment and refund information.

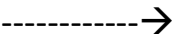
Column	Type	Constraints	Purpose
transaction_id	INT	PK, AUTO_INCREMENT, NOT NULL	Unique transaction ID
Booking_id	INT	FK → BOOKINGS, NOT NULL	Related booking
transaction_type	VARCHAR (20)	NOT NULL	PAYMENT/ REFUND
payment_method	VARCHAR (30)	NULL	UPI/ CARD/ NET_BANKING
amount	DECIMAL (10,2)	NOT NULL	Transaction amount
transaction_date	DATETIME	DEFAULT CURRENT_TIMESTAMP	Transaction date/time
status	VARCHAR (20)	NOT NULL	SUCCESS/ FAILED/ PENDING
reason	VARCHAR (255)	NULL	Reason for refund/failure etc.

SCHEMA DESIGN:



MOVIES
PK movie_id
title
language
genre
duration_minutes
release_date
theatre_name
location
show_date
start_time
end_time
ticket_price
status
FK created_by
created_at

1:N



SEATS
PK seat_id
FK movie_id
seat_number
seat_type
status
FK locked_by
locked_at
expires_at

Relationship Between Tables:

Relationship	Type	Meaning
USERS → BOOKINGS	1 : N	One customer can make many bookings
USERS → SEATS	1 : N	One user can temporarily lock seats
USERS → MOVIES	1 : N	Admin/provider can create many movies
MOVIES → SEATS	1 : N	One movie/show can have many seats
SEATS → BOOKINGS	1 : N	A seat can appear in bookings over different show/movie records
BOOKINGS → TRANSACTIONS	1 : N	One booking can have payment/refund transactions